Prop: A 2×2 matrix in $\mathbb{R}$ has its determinant equal to the product of its eigenvalues.

Proof: Given a matrix M $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, finding the determinant is equal to finding the root of the polynomial $\det\begin{pmatrix} a-x & b \\ c & d-x \end{pmatrix}$

$= (a-x)(d-x) - bc = x^2 - (a+d)x + (ad-bc)$

Let λ_1, λ_2 be the roots of the polynomial (i.e. the eigenvalues).

So $(x-\lambda_1)(x-\lambda_2) = x^2 - (\lambda_1+\lambda_2)x + \lambda_1\lambda_2$

So $\lambda_1\lambda_2 = ad - bc$ as required.

Prop If $(h\ k)\begin{pmatrix} L_p & M_p \\ M_p & N_p \end{pmatrix}\begin{pmatrix} h \\ k \end{pmatrix} > 0$ for all $h, k \in \mathbb{R}$.

then $\det\begin{pmatrix} L & M \\ M & N \end{pmatrix} = LN - M^2 > 0$.

Proof: $\begin{pmatrix} L_p & M_p \\ M_p & N_p \end{pmatrix}$ is symmetric so all its eigenvalues are real. (LA 1)

For such an eigenvalue λ, find its non-zero eigenvector $\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, consider

$$(v_1\ v_2)\begin{pmatrix} L_p & M_p \\ M_p & N_p \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = (v_1\ v_2)\,\lambda\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \lambda(v_1^2 + v_2^2).$$

But $(v_1\ v_2)\begin{pmatrix} L_p & M_p \\ M_p & N_p \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} > 0$ by assumption,

so $\lambda(v_1^2 + v_2^2) > 0$, the second term must be positive,

so λ can't be negative or zero, (i.e. it is positive)

So all of $\begin{pmatrix} L_p & M_p \\ M_p & N_p \end{pmatrix}$'s eigenvalue are positive, so $\det\begin{pmatrix} L_p & M_p \\ M_p & N_p \end{pmatrix}$ $= L_pN_p - M_p^2 > 0$ by above Lemma.

Remark. In this calculation, everything is specific to the "local minimum" point $\vec{p} = (x_0, y_0, z_0)$ given.